# How Many Labels Before Which Labels? Risk-Budgeted Feedback Acquisition under Selective Labels

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**Abstract.** Lending decisions determine which repayment outcomes become observable and hence which data are available for future learning. Existing work studies exploration under selective feedback, but an information-oriented approval rule can reduce the number of outcomes acquired under a risk constraint. We study this trade-off with lexicographic risk-budgeted feedback acquisition: at each policy state, first maximize the feasible label count under predicted applicant-level loss, then optimize a secondary score at that count. We compare policies in random-order offline replay across four credit datasets, five budget ratios, and separate development and confirmation seeds. At $\rho = 0.1$, LRBE-FAVE acquires 67.1 more labels than QualityFirst-FAVE on HMEQ and 209.6 more on LendingClub. When quantity is preserved, FAVE uses more predicted risk than minimum-risk selection without a consistent cross-dataset Recall or Brier advantage. These whole-policy comparisons motivate a quantity-preserving reference for evaluating acquisition scores; they do not identify a causal effect of feedback composition. We also analyze raw feedback gains, rather than unstable reference-gap ratios, and distinguish predicted feasibility from ex-post realized-loss exceedance.

**Keywords:** Selective labels · Feedback acquisition · Risk budgets · Credit prediction

## 1 Introduction

Credit decisions determine which data become available for subsequent learning. Repayment outcomes are observed for approved applicants, while the outcomes of rejected applications remain unknown. This selective-label problem can leave evaluation and retraining dependent on the policy that collected the data [14,13]. Acquiring feedback from a previously rejected region therefore entails a lending decision, not simply an annotation request: observing an outcome requires accepting the associated credit exposure.

Existing research addresses this dependence through reject inference [4], decision policies for selective observations [22], optimistic exploration [19], and adaptive bounded debiasing [24]. Recent credit experiments also use controlled approvals to investigate feedback loops [20]. These approaches motivate a practical question about resource allocation: under a predicted-loss budget, does

sacrificing feasible feedback quantity for a higher information score deliver sufficient downstream benefit? Non-uniform acquisition costs are already studied in active learning [8,21]; our focus is the comparison of quantity-preserving and composition-oriented policies when the cost is an applicant-level credit-loss proxy.

We examine this question using lexicographic risk-budgeted feedback acquisition (LRBE). At each round, the first stage computes the maximum feasible count $K_t^*$ under predicted costs $\widehat{c}_{it} = \widehat{p}_{it}\mathrm{EAD}_i\mathrm{LGD}_i$. The second stage optimizes a sample-composition score subject to that count equality. This structural constraint makes quantity sacrifice observable: a secondary score can change which applicants are selected but cannot reduce the current maximum count. We compare this rule with minimum-risk, composition-first, joint-objective, and randomized selection. The contribution is the resulting strategy comparison and its empirical consequences, not a new solution to the maximum-count optimization problem.

The evaluation uses four credit datasets and five budget ratios in random-order sequential offline replay. Policies start from the same initial data and incoming batches, then retrain on their own acquired outcomes. Their later candidate sets and numerical budgets can therefore differ. Paired comparisons measure whole-policy outcomes under shared exogenous conditions, rather than independent causal effects of quantity, composition, and risk. Across these experiments, composition-first selection often acquires substantially fewer labels. FAVE and minimum-risk selection have similar observed cumulative counts, but FAVE generally uses more predicted risk; we do not observe a consistent cross-dataset downstream advantage for the more complex policy. Dataset-specific improvements remain part of the evidence.

Our contributions are threefold. First, LRBE makes the trade-off between quantity and composition explicit within each policy state by separating a count-maximizing stage from a second-stage composition score. Second, across four datasets, policies allowed to give up current feasible count often obtain fewer labels without a consistent downstream benefit. FAVE also uses additional predicted risk without a stable advantage over MinRisk. Third, raw feedback-gain analyses and realized-loss exceedance analyses distinguish intermediate feedback diagnostics, predictive outcomes, and ex-post loss. Together, these results support MinRisk as a quantity-preserving reference before crediting a more complex composition objective with downstream value.

## 2 Related Work

*Learning from selectively observed outcomes.* Work on selective labels studies the evaluation and learning problems created by decision-dependent observation [14,13]. Expert consistency and stochastic fair decisions provide ways to learn or collect data beyond existing acceptance patterns [6,12]. Credit reject inference examines augmentation and sample-selection correction [4,16], while finite-domain policy optimization and one-sided-feedback learning formalize se-

quential decisions with missing outcomes [22,10]. Our replay retains observed historical outcomes and reveals them according to simulated approvals. It evaluates an acquisition policy, not an imputation method or a field trial on actual rejected borrowers.

*Exploration objectives and costs.* PLOT uses optimistic pseudo-labels to counter self-confirming rejection and gives regret guarantees under stated modeling assumptions [19]. Adaptive bounded exploration trades debiasing against the risk of accepting otherwise rejected individuals [24]. Partial-feedback exploration also supports subgroup coverage and fairness objectives [11], while cost-efficient fairness auditing accounts for assessment, processing, and potential-default costs [5]. Controlled approvals in credit scoring examine the consequences of retraining under selection bias [20]. These are direct precedents for costly feedback acquisition. We instead compare how a current predicted-loss allowance is allocated between feedback quantity and composition. The randomized controls in our experiments are in-house strategy controls, not implementations of these published algorithms.

*Resource constraints and predictive evaluation.* Cost-sensitive active learning already accounts for heterogeneous query costs [8,21]. Contextual bandits with knapsack constraints and conservative exploration place learning within resource or baseline-performance restrictions [1,23]; safety-constrained approaches further consider uncertainty in feasibility [2,18]. LRBE uses a point predicted-loss constraint rather than these sequential guarantees. Its secondary scores are evaluated by the resulting data and predictions, not by score values alone. Credit benchmarking and probability-estimation research motivate reporting both discrimination and probability error [15,17]. Accordingly, we compare Recall and Brier score while reporting predicted-budget compliance separately from realized defaulted exposure.

## 3 Risk-Budgeted Feedback Acquisition

### 3.1 Sequential observation and policy-specific budgets

At round $t$, policy $m$ receives an incoming batch $\mathcal{U}_t$ and fits a default predictor on its visible labeled set $\mathcal{L}_t^{(m)}$. An outcome $y_i = 1$ denotes default. Ordinary approvals have predicted default probability below the fixed threshold $\tau = 0.5$, which serves as a common operating point for every policy in the replay. The remaining applicants form the exploration candidates

$$\mathcal{C}_t^{(m)} = \{i \in \mathcal{U}_t : \widehat{p}_{it}^{(m)} \geq \tau\}. \tag{1}$$

The policy selects exploratory approvals $\mathcal{S}_t^{(m)} \subseteq \mathcal{C}_t^{(m)}$. Only outcomes of ordinary and exploratory approvals are added to the next visible set. Rejected applicants are not carried over to a backlog. Outcomes are revealed within the simulation round; actual repayment delays are not modeled.

Write $a_i$ for exposure and set $\mathrm{LGD}_i = 1$. The planning cost, exploration budget, and selected risk are

$$\widehat{c}_{it}^{(m)} = \widehat{p}_{it}^{(m)} a_i, \qquad B_t^{(m)} = \rho \sum_{i \in \mathcal{C}_t^{(m)}} \widehat{c}_{it}^{(m)}, \qquad \widehat{R}_t^{(m)} = \sum_{i \in \mathcal{S}_t^{(m)}} \widehat{c}_{it}^{(m)}. \tag{2}$$

The constraint $\widehat{R}_t^{(m)} \le B_t^{(m)}$ covers exploratory approvals only, not the entire lending portfolio. Defaulted exposure is evaluated separately as $R_t^{\mathrm{real},(m)} = \sum_{i \in \mathcal{S}_t^{(m)}} y_i a_i$. With unit LGD this is an experimental loss proxy, not recovery-adjusted net loss or full IFRS 9 accounting.

Policies share the initial split and exogenous batches, but their feedback choices affect subsequent training sets, predictions, candidates and budgets. Pairing on dataset, seed and $\rho$ therefore compares complete policies under common incoming data. It does not fix their subsequent numerical budgets or isolate a causal effect of composition. Superscripts $m$ are suppressed below for within-policy optimization.

### 3.2 Lexicographic quantity and composition

LRBE first determines the maximum feasible exploratory count:

$$K_t^* = \max_{z \in \{0,1\}^{|\mathcal{C}_t|}} \sum_i z_i \quad \text{subject to} \quad \sum_i z_i \widehat{c}_{it} \le B_t. \tag{3}$$

For nonnegative costs and unit count value, sorting costs and taking the longest feasible prefix solves this stage in $O(n \log n)$ time. This elementary cardinality construction is used as a reference, not claimed as a new optimization algorithm. A fixed, outcome-blind secondary score $q_i$ then determines the composition:

$$\max_z \sum_i z_i q_i \quad \text{subject to} \quad \sum_i z_i = K_t^*, \quad \sum_i z_i \widehat{c}_{it} \le B_t, \quad z_i \in \{0,1\}. \tag{4}$$

The equality prevents the secondary objective from sacrificing feasible count *at the current policy state*. It does not force distinct policies to acquire identical cumulative counts after their histories diverge. Both stages use only currently available features and predictions, never unobserved candidate outcomes or test labels.

### 3.3 Acquisition policies

**CountOnly-MinRisk** minimizes predicted risk among maximum-count sets by setting $q_i = -\widehat{c}_i$. It is abbreviated MinRisk in tables. **LRBE-Uncertainty** uses $q_i = \widehat{p}_i(1 - \widehat{p}_i)$. **LRBE-FAVE** uses the frozen FAVE-v2.1 score, a feature-space local information proxy (Appendix A). Its Fisher-style geometry is a surrogate constructed from predictions and covariates, not the Hessian of the fitted boosted-tree model or a guarantee of retraining improvement.

**Table 1.** Exact input cohorts and exposure semantics. Raw risk units are not comparable across datasets.

| Dataset | Records | Exposure and evidence role |
|---|---|---|
| HMEQ | 5,960 | LOAN amount; development |
| LendingClub | 16,566 | loan-amount proxy; external validation |
| UCI | 30,000 | unit exposure; development |
| PPDai | 55,596 | unit exposure; exploratory external |

**Quality-first policies** maximize the corresponding nonnegative uncertainty or FAVE utility under the risk constraint without the count equality. **Joint-Best-Val-Uncertainty** maximizes $\sum_i z_i(\bar{q}_i - \lambda \bar{c}_i)$ over positive-score candidates, where bars denote within-pool min–max normalization. Its parameter is chosen using only the initial visible set, split into 60% training, 20% pseudo-candidates and 20% validation. Validation PR-AUC determines the choice, with Recall and then smaller $\lambda$ as tie-breakers. These are diagnostic comparators for the acquisition objectives, not implementations of all prior exploration algorithms.

Two additional controls vary randomization. **RiskBounded-Random** visits candidates in a reproducible pseudo-random order and accepts each if its cost fits the remaining budget. **RiskBounded-Epsilon** instead preserves $K_t^*$ and uses $q_i = -0.9\bar{c}_i + 0.1u_i$, where $u_i$ is a deterministic hash-based random score. We retain its archive name but call it the *random-score* control in prose: 0.1 is an objective weight, not a 10% random-approval probability or conventional epsilon-greedy rule. **NoExplore** reveals ordinary approvals only. **Oracle** also reveals every candidate outcome and ignores the exploration budget; it is a full-feedback diagnostic, not a guaranteed upper bound on test performance.

## 4 Experimental Design

### 4.1 Data and replay protocol

For each seed, a random permutation assigns 25% of records to a fixed outer test set. The first 20% of the remaining stream form the initial visible set, with a minimum of 100 records; all remaining records are split into ten incoming batches. The initial visible set is therefore a random cohort sample rather than the output of a historical approval rule. Policy-dependent selective observation begins only when the policies process the ten subsequent batches: ordinary approvals are observed automatically, whereas exploratory approvals are chosen from the threshold-defined candidate set. Thus the experiment is *random-order offline replay*, not historical-time evaluation.

Fresh confirmation seeds 1001–1010 are disjoint from development seeds 0–19. The fresh design includes an objective comparison at $\rho = 0.1$, a four-policy sweep over $\rho \in \{0.025, 0.05, 0.1, 0.2, 0.4\}$, and NoExplore/Oracle references. The two randomization controls use the same fresh seeds at $\rho = 0.1$. Development

analyses are retained separately because their protocols differ from the fresh uncertainty estimates. All reported analyses use the registered seeds, policies, and archived raw outputs.

### 4.2 Model, optimization and observation controls

All fresh policies use XGBoost with 100 trees, maximum depth 3, learning rate 0.1, binary logistic objective, and histogram-based training. Numeric features receive median imputation and standardization; categorical features receive a missing-value category and one-hot encoding. Target, ID and the designated exposure column are excluded from predictors. Preprocessing is fitted on visible records only. Each round fits a pre-acquisition model for selection and a post-acquisition model for test evaluation. The outer test set never selects parameters or candidates; population default rates and selected outcomes are used only in offline diagnostics.

Maximum-count selection uses a sorted-cost prefix with a $10^{-10}$ numerical allowance. The common second-stage binary program and quality-first/joint programs use SciPy's mixed-integer solver; an unsuccessful solve raises an error. Risk auditing permits a $10^{-6}$ absolute allowance. Candidate costs are strictly positive under the validated exposures and rejection threshold, so zero budgets yield zero feasible count. Empty candidate sets also yield no exploratory selections. GPU training and supported FAVE matrix operations accelerate computation, while optimization remains CPU-side. Appendix A records implementation and validation details.

### 4.3 Outcomes and statistical comparisons

We report cumulative exploratory count, cumulative predicted risk, final-test default Recall, and Brier score. Recall is $TP/(TP + FN)$ at $\tau = 0.5$; Brier is the mean squared error of default probabilities. The acquired default fraction describes the selected outcomes. A separate feedback diagnostic, $F$, is the default fraction in the entire visible set divided by the default fraction in the non-test stream. It therefore includes ordinary approvals and the initial set, not only exploratory selections. Neither $F$ nor mean selected predicted probability establishes exact applicant overlap or an independent composition effect.

At each dataset and budget ratio, final policy differences are paired over seeds. Tables show mean differences and pointwise 95% percentile bootstrap intervals from 10,000 seed resamples. Counts that are close descriptively are not called experimentally matched; intervals crossing zero are not equivalence or noninferiority tests. The comparisons are exploratory across multiple metrics and conditions, so isolated interval exclusions are not familywise significance claims.

For feedback–prediction associations, the primary variables are raw gains from the same-seed NoExplore reference: $\Delta F = F_m - F_N$, $\Delta\text{Recall} = \text{Recall}_m - \text{Recall}_N$, and $\Delta\text{Brier} = \text{Brier}_N - \text{Brier}_m$. Positive Brier gain means improvement.

Spearman associations are computed separately within dataset and budget, retaining all methods belonging to a resampled seed in each of 10,000 cluster-bootstrap draws. Budget-centered associations are secondary descriptions, not causal estimates. Reference-gap ratios are supplemental only: $(M_m - M_N)/(M_O - M_N)$ is interpretable as recovery only when the oriented Oracle gap is positive; nonpositive or numerically unresolved gaps are N/A. The primary raw-gain analysis avoids dividing by these unstable reference differences.

*Cohort provenance.* UCI is the Default of Credit Card Clients dataset [25,26]; HMEQ is the home-equity cohort documented in [3]. LendingClub uses a mature Fully Paid/Charged Off cohort from the DePaul-hosted 2015 sample [7]. Its outcomes concern historically issued loans, not observed outcomes for genuinely rejected applicants. PPDai is linked to Figshare file 26049578 [9], with incomplete item-level and feature-semantic provenance, so it carries lower evidentiary weight. Its exact processed exposure column is identically one. Hash-checked source instructions identify the exact inputs without redistributing third-party data. The strict exposure audit finds no missing, nonnumeric, nonfinite or nonpositive exposure in the three stored exposure columns; UCI constructs unit exposure directly.

## 5 Results

### 5.1 The cost of sacrificing current feasible count

The central empirical question is whether a secondary acquisition objective earns the feedback quantity it gives up. Removing the count equality quantifies that cost. Relative to QualityFirst-FAVE, LRBE-FAVE acquires 67.1 [62.6, 71.4] additional labels on HMEQ, 209.6 [202.5, 216.1] on LendingClub, 2.0 [0.9, 3.1] on PPDai, and 56.4 [54.7, 58.2] on UCI. Recall favors LRBE-FAVE on HMEQ by 0.0175 [0.0041, 0.0297]; the other three Recall intervals contain zero. The quantity ordering follows from the current-state count constraint. The empirical result is that giving up this quantity does not yield a consistent predictive improvement across the four datasets.

The random-order control provides a second view of the same trade-off. Against MinRisk, it acquires 87.0 fewer labels on HMEQ, 317.6 fewer on LendingClub, and 59.5 fewer on UCI; PPDai differs by only 0.3. HMEQ Recall is lower by 0.0268 [-0.0395, -0.0140], while the other Recall intervals contain zero. By contrast, the fixed-count random-score control remains close in cumulative count, although later policy states still differ: $+1.3$ labels on HMEQ, $-1.5$ on LendingClub, $+0.3$ on PPDai, and $-2.7$ on UCI. Its Recall and Brier intervals against MinRisk contain zero in every dataset. These controls show how current-state cardinality preservation changes a complete policy; they do not isolate cardinality from all later state changes.

Table 2 then asks whether a more elaborate score adds value once the count constraint is retained. FAVE and MinRisk have close observed cumulative counts,

**Table 2.** Fresh paired differences, FAVE minus MinRisk, at $\rho = 0.1$ ($n = 10$ seeds). Brackets are pointwise 95% bootstrap intervals. Negative $\Delta$Brier favors FAVE.

| Data | $\Delta K$ | $\Delta \widehat{R}$ | $\Delta$Recall | $\Delta$Brier |
|---|---|---|---|---|
| HMEQ | 1.1 | 32,605.0 | -0.0007 | 0.00018 |
| | [-0.9,3.0] | [22,925,43,035] | [-.0083,.0092] | [-.00012,.00056] |
| LendingClub | -0.1 | 31,548.8 | 0.0000 | 0.00016 |
| | [-2.5,2.6] | [26,823,35,982] | [-.0039,.0042] | [-.00003,.00037] |
| PPDai | 0.0 | 0.68 | -0.0002 | -0.00002 |
| | [-1.0,1.1] | [.13,1.27] | [-.0028,.0023] | [-.00006,.00002] |
| UCI | -0.3 | 2.16 | -0.0004 | -0.00018 |
| | [-2.0,1.5] | [1.36,2.98] | [-.0033,.0029] | [-.00028,-.00008] |

but this is an empirical similarity rather than a matching constraint across policies.

FAVE uses more predicted risk in all four datasets, yet every Recall interval contains zero. Three of the four Brier intervals contain zero; the exception is the small UCI improvement shown in the table. Thus the results retain a dataset-specific benefit, but they do not show a consistent cross-dataset return to FAVE's additional predicted risk.

### 5.2 Budget sensitivity

The budget ratio has a much larger and more consistent relation with acquired quantity than the secondary score. HMEQ MinRisk grows from 57.4 labels at $\rho = 0.025$ to 374.0 at $\rho = 0.4$; FAVE grows from 58.0 to 375.8. LendingClub MinRisk grows from 207.6 to 1,253.1 and FAVE from 206.3 to 1,253.3. Recall moves less and method ordering changes across budgets. Figure 1 shows representative HMEQ and LendingClub trajectories. The sweep changes both the scale of feedback and the subsequent policy state, so it is a sensitivity analysis rather than a quantity-only intervention.

### 5.3 Feedback diagnostics do not have a single translation pattern

Normalized recovery divides policy changes by an Oracle–NoExplore gap, and is not interpretable as recovery when Oracle is not better. In the present data, the Recall gap is nonpositive in 10 of 40 dataset–seed references: two LendingClub seeds, two PPDai seeds, including one zero, and six UCI seeds. Oracle gaps are also nonpositive in 21 of 40 ROC-AUC and 15 of 40 PR-AUC references. We therefore use raw gains from NoExplore as the primary analysis and reserve normalized recovery for positive-gap cells.

Figure 2 relates visible-set default-ratio gain to Recall and Brier improvement at $\rho = 0.1$. HMEQ has a positive Recall association, $\rho_s = 0.397$ [0.013, 0.660], but its Brier association is uncertain, 0.062 [−0.175, 0.381]. LendingClub, PPDai, and UCI Recall intervals contain zero at this budget, as do all four Brier intervals.

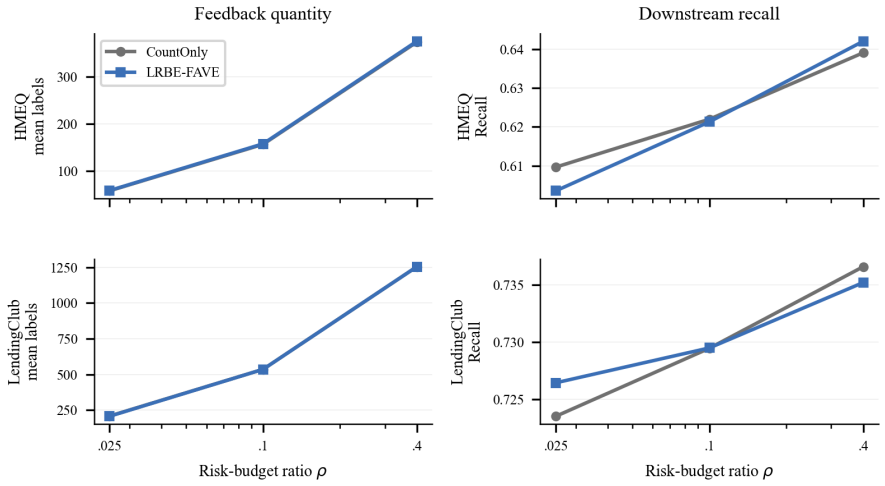



**Fig. 1.** Mean cumulative exploratory count and final Recall across budget ratios for HMEQ and LendingClub (ten fresh seeds). Larger budgets increase feedback, while MinRisk and FAVE do not have a uniform Recall ordering.

Across the five within-budget analyses, HMEQ Recall estimates range from 0.389 to 0.514 and all intervals exclude zero. LendingClub instead has negative Recall associations at $\rho = 0.025$ and 0.05, followed by uncertain estimates. PPDai's smallest-budget cells have near-degenerate ranks and yield perfect coefficients, so those boundary values are not interpreted as general evidence.

These associations concern a diagnostic of the complete visible set, which includes the initial set and ordinary approvals. A larger ratio is not necessarily closer to the population target, and policy-specific predictors make selected mean PD an unsuitable cross-policy identity measure. The supported conclusion is consequently heterogeneous translation: feedback diagnostics can align with Recall in some regimes but are not a stable proxy across datasets, budgets and outcomes. The correlations do not estimate a causal composition effect.

### 5.4 Predicted feasibility and realized default exposure

All non-Oracle selections obey the fitted point-cost constraint within audit tolerance. Across 104 equally sized method–dataset–budget groups, the mean rate at which realized defaulted exposure exceeds the policy's numerical predicted budget is 0.4798, with group rates ranging from 0.01 to 0.84. Because each group contains 100 seed–round observations, the unweighted group mean also equals the observation-weighted mean for these included cells. The range is descriptive, not a confidence interval.

This comparison evaluates an ex-post loss exceedance against a budget constructed from fitted expected costs, rather than against a realized-loss quantile.

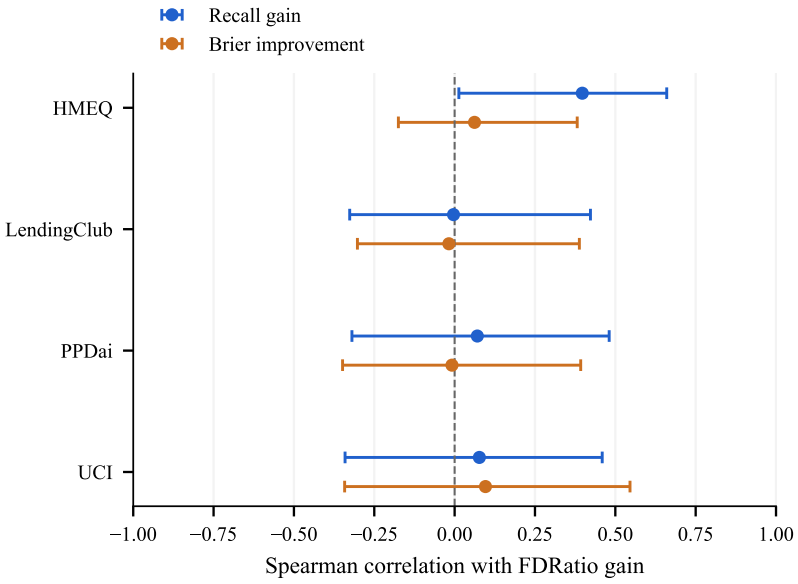



**Fig. 2.** Within-dataset associations between visible-set default-ratio gain and downstream improvement at $\rho = 0.1$. Points use 40 terminal policy cells (four policies, ten seeds); bars are pointwise 95% intervals from 10,000 whole-seed bootstrap resamples. Positive Brier gain means lower error.

The exceedance rate therefore shows that point-estimate feasibility does not ensure ex-post loss-cap compliance. It does not separately identify model miscalibration, stochastic default variation, exposure assumptions, or dependence among loans.

*Descriptive computation time.* Recorded episode wall time includes model fitting, selection, evaluation and concurrent system effects. Median LendingClub times are about 44.2 s for MinRisk and 80.1 s for FAVE in the objective-comparison block; HMEQ, UCI and PPDai medians differ by less than 0.1 s between these two policies. The LendingClub difference flags a possible computational cost but is not an isolated acquisition-latency benchmark.

## 6 Discussion and Limitations

*What a quantity-preserving reference contributes.* Maximizing feasible count is useful here because it makes the opportunity cost of a secondary acquisition score observable. The relevant question is not whether the count-maximizing rule selects more applicants at a fixed state, which follows from its definition, but whether policies that sacrifice count obtain enough predictive benefit to justify

that choice. The comparisons show substantial quantity differences without a consistent cross-dataset advantage for composition-first selection. MinRisk is consequently a useful reference for this protocol. This is not a prescription to approve as many applicants as possible outside the eligibility rule and predicted-risk constraint.

*Policy effects and composition evidence.* Similar average exploratory counts do not control feedback quantity experimentally. Ordinary approvals, visible training sets, fitted costs and available budgets can all evolve differently. The observed FAVE–MinRisk differences therefore compare complete acquisition policies, not the marginal value of composition at equal count and equal risk. Mean selected PD is also evaluated under each policy's own predictor. A change in this summary cannot by itself identify which applicants were exchanged or why a prediction improved. Raw feedback–prediction associations describe joint variation, while the direct policy contrasts assess whether the more complex rule delivers measurable predictive gains. Neither analysis should substitute for the other.

*Scope of inference.* The replay is conditional on historical cohorts with observed outcomes. In particular, LendingClub does not provide repayment outcomes for actual historical rejects. Random ordering, an initially sampled visible set, immediate label revelation, a single model family, and the fixed operating threshold $\tau = 0.5$ define the evaluated setting. The threshold makes the policies comparable within the replay; it is not intended to represent a production credit cutoff. The results therefore do not establish historical-time robustness or benefits under delayed feedback. PPDai provides unit-exposure exploratory evidence, not a second validation of monetary exposure heterogeneity. The comparison also does not include full implementations of the closest online exploration algorithms. These boundaries narrow the empirical claim rather than make the unevaluated comparisons unnecessary.

*Risk interpretation.* An optimizer can enforce a sum of predicted losses without controlling the tail of realized loss. Here, an exceedance means that realized defaulted exposure in a seed–round observation is larger than the policy's point-estimate budget. It is an ex-post comparison with an expected-loss quantity, not a direct estimate of safety failure. The observed frequency does not attribute the difference separately to probability error, default randomness, or exposure assumptions. Live credit experimentation would require product-specific exposure and recovery definitions, delayed-outcome handling, concentration and fairness safeguards, and explicit stopping rules. None of these operational guarantees follows from the current point-cost constraint.

## 7 Conclusion

This study evaluates how feedback quantity and sample-composition objectives interact in risk-budgeted selective-label learning. LRBE preserves the current

maximum feasible exploratory count before optimizing a secondary score. In the evaluated offline policies, composition-first or random-order acquisition can forgo substantial feedback, while FAVE does not provide a consistent cross-dataset Recall or Brier advantage over minimum-risk selection despite using more predicted risk. Feedback diagnostics and ex-post defaulted exposure provide complementary evidence, not substitutes for predictive evaluation or statistical safety guarantees. The resulting recommendation is methodological: report quantity, predicted-risk use, and downstream performance together when assessing whether an acquisition score earns its cost.

## A Secondary Score and Reproducibility

*Frozen FAVE-v2.1 score.* Let $v_j = [1, x_j^\top]^\top$ be the augmented, preprocessed feature vector. For the current visible set $\mathcal{L}_t$ and label-free reference batch $\mathcal{U}_t$, define

$$H_t = \sum_{j \in \mathcal{L}_t} \max\{\widehat{p}_j(1-\widehat{p}_j), 10^{-6}\} v_j v_j^\top, \tag{5}$$

$$G_t = |\mathcal{U}_t|^{-1} \sum_{j \in \mathcal{U}_t} \widehat{p}_j v_j v_j^\top, \qquad \eta = \max\{10^{-6}\,\mathrm{tr}(H_t)/d, 10^{-10}\}, \tag{6}$$

$$A_t = H_t + \eta I, \qquad \widetilde{w}_i = \max\{\widehat{p}_i(1-\widehat{p}_i)\widehat{p}_i^{0.5}, 10^{-10}\}, \tag{7}$$

$$q_i^{\mathrm{FAVE}} = \frac{\widetilde{w}_i v_i^\top A_t^{-1} G_t A_t^{-1} v_i}{1 + \widetilde{w}_i v_i^\top A_t^{-1} v_i}. \tag{8}$$

Here $d$ includes the intercept coordinate. Predicted probabilities are clipped to $[10^{-9}, 1-10^{-9}]$ before these operations; nonfinite score outputs are zeroed and scores are clipped below at zero. The default-weighting exponents and numerical regularization are fixed. This linearized feature-space construction is used with boosted-tree predictions as a scoring proxy; it is not an exact influence calculation for the tree-fitting procedure. The Joint comparator selects $\lambda \in \{0.01, 0.1, 0.5, 1, 2, 5, 10\}$ using only initial-set validation. The initial split is stratified where possible, with a deterministic random split as fallback.

*Execution archive and analysis.* The fresh archive contains 1,120 run files (240 objective-comparison, 800 budget-sweep and 80 reference files), each with an initial row and ten post-acquisition rows. The randomized block adds 80 files. Overlapping settings in separate blocks are not independent replications and are never pooled to increase sample size. The fresh primary comparisons have ten paired seeds; older 20-seed evidence retains its own analysis identity. Source hashes, full-precision tables, parameter grids, and environment records accompany the executable analysis. The publication tables use pointwise seed-bootstrap intervals and retain negative or uncertain differences.

*Input and trajectory audit.* Exact-input SHA-256 checks cover HMEQ, LendingClub and PPDai. All 78,122 records have numeric, finite, positive exposure;

PPDai values are all one. UCI constructs ones directly. The strict input loader rejects invalid exposure rather than replacing it by zero. Eight validation trajectories (four datasets, seed 1001, MinRisk and FAVE) agree with the archived outputs in the checked counts, risk values, and predictive metrics. This check is not an additional statistical replication. Of 40 paired post-initial rounds, incoming-batch hashes agree in all 40, while candidates and numerical budgets agree in only four. This directly documents the policy-specific state evolution.

*Validation coverage.* Tests cover exposure rejection, required parity fields, round keys, nonfinite-value handling, recovery restricted to positive reference gaps, and prior split/selection invariants. A parity check fails on a missing required metric rather than silently omitting it. The analysis records invalid reference-gap counts and leaves raw experiment files untouched. Data hashes establish exact file identity, not unbiased cohort selection; executable checks establish their stated invariants, not the absence of every implementation defect.

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